Revisiting Isolated DC-DC Converters-II

Introduction

In the realm of modern power electronics, particularly for applications like charging infrastructure, isolated DC-DC converters are indispensable. They provide galvanic isolation between the input and output, crucial for safety and noise reduction. This lecture delves deeper into the intricacies of these converters, building upon previous discussions of the basic forward converter. While the forward converter offers a fundamental approach to isolated power conversion, it comes with several inherent limitations that restrict its performance and efficiency. Understanding these constraints is vital for designing robust and optimized power systems. This session explores various modifications to the forward converter topology aimed at addressing these limitations and introduces the push-pull converter, a topology designed to significantly enhance transformer core utilization, thereby improving overall converter efficiency and compactness.

Core Concepts

Term	Definition	Significance
`Forward Converter`	A DC-DC converter that transfers energy from input to output when the primary switch is ON, using a transformer for isolation.	Serves as a basic isolated topology but has limitations in core utilization and switching frequency.
`Demagnetizing Winding`	An auxiliary winding on the transformer, connected with a diode, used to reset the core flux during the switch OFF period.	Enables core reset in forward converters, but adds complexity and can increase switch voltage stress.
`Dual Switch Forward Converter`	A forward converter topology employing two primary switches and two diodes to achieve lossless core reset.	Offers lossless core reset and eliminates voltage kicks from leakage inductance, but still underutilizes the core.

Term	Definition	Significance
`Push-Pull Converter`	An isolated DC-DC converter using a center-tapped primary winding and two switches to drive the transformer in both directions.	Achieves full core utilization by swinging flux from +Bmax to -Bmax, leading to smaller transformer size.
`Core Utilization`	The extent to which the magnetic core material is effectively used, typically measured by the flux density swing.	Higher utilization allows for smaller, lighter, and more efficient transformers, reducing overall converter size and cost.
`Flux Walking`	A phenomenon in push-pull converters where unequal voltage-seconds applied to the transformer windings cause a DC offset in the flux.	Can lead to core saturation over time, severely damaging the converter if not mitigated by control strategies.
`Leakage Inductance`	The portion of the transformer's inductance that does not couple magnetic flux between primary and secondary windings.	Causes voltage spikes (kicks) on switching devices during turn-off, requiring higher voltage ratings for switches.
`Reverse Recovery Losses`	Energy dissipated in a diode when it transitions from forward conduction to reverse blocking, due to stored charge.	A significant source of power loss, especially at higher switching frequencies, impacting overall converter efficiency.
`Duty Ratio (D)`	The fraction of the switching period during which a switch is ON, controlling the output voltage.	Limited in forward converters by the need for sufficient core reset time, restricting operating range and frequency.
`Switching Frequency (f_s)`		Higher frequencies allow for smaller passive components

Term	Definition	Significance
	The rate at which the power switches in the converter are turned ON and OFF.	but can exacerbate losses like reverse recovery and core reset issues.

Detailed Analysis

1. Limitations of the Basic Forward Converter

The fundamental forward converter, while simple, presents several operational constraints that limit its performance:
* **Limited Duty Ratio and Switching Frequency**: The transformer core requires a sufficient reset time during the switch OFF period, typically `(1-D)Ts`. This necessity restricts the maximum allowable duty ratio (D) and, consequently, the switching frequency. If the frequency is too high, `Ts` becomes small, and `(1-D)Ts` might be insufficient for complete flux reset, leading to core saturation.
* **Significant Reverse Recovery Losses**: Diodes in the secondary side switch in every cycle, leading to considerable reverse recovery losses. These losses contribute to the overall inefficiency of the converter, especially at higher switching frequencies.
* **Lossy Core Reset**: The core reset mechanism often involves a reset winding and a resistor (`Rf`), resulting in power dissipation across `Rf`. This makes the reset process lossy, further reducing efficiency.
* **Limited Core Utilization**: The flux in the transformer core typically swings from zero to a maximum positive value (`Bmax`) and then back to zero. It does not enter the negative flux density region. This means only about half of the core's magnetic capability is utilized, requiring a larger core than theoretically necessary for a given power level. For instance, the area of the core is proportional to `V / (N * Bmax * f)`, whereas full utilization would allow `V / (2 * N * Bmax * f)`.
* **High Switch Voltage Rating**: The primary switch must block a voltage higher than the input voltage, often `Vin + (Np/Nd)*Vin` (where `Nd` is the demagnetizing winding turns), which can be substantial.

2. Forward Converter with Demagnetizing Winding

To address the lossy reset and improve core utilization, a common modification is the forward converter with a demagnetizing winding.
* **Topology**: This configuration uses three windings: a primary winding (`Np`), a secondary winding (`Ns`), and a demagnetizing winding (`Nd`) connected with a diode (`DS1`).

* **Operation during `DTs` (Switch ON)**:
* The primary switch (S1) is ON.
* Voltage `Vin` is applied across `Np`.
* Secondary voltage `(Ns/Np)*Vin` is induced, forward-biasing the output diode (D1) and reverse-biasing D2.
* The demagnetizing winding diode (`DS1`) is reverse-biased.
* Flux in the core rises linearly with a slope of `Vin/Np`.
* The output inductor current (`IL`) rises.
* **Operation during `(1-D)Ts` (Switch OFF)**:
* S1 is OFF.
* The output inductor current flows through D2 (forward-biased), and D1 becomes reverse-biased.
* The demagnetizing winding diode (`DS1`) becomes forward-biased due to the collapsing flux.
* The demagnetizing winding applies a voltage across the core, causing the flux to fall linearly with a slope of `-Vin/Nd`. This actively resets the core.
* **Advantages**: Provides an active, less lossy core reset compared to a simple resistor.
* **Disadvantages**:
* Requires a multi-winding transformer, increasing complexity.
* Still has limited switching frequency because sufficient time must be provided for the flux to reset to zero.
* The switch S1 must block a voltage of `Vin + (Np/Nd)*Vin`, which can be higher than the input voltage. A trade-off exists: increasing `Nd` to speed up reset (steeper slope) also increases the switch voltage rating.
* **Voltage Kick due to Leakage Inductance**: When S1 turns OFF, the current flowing through the primary winding's leakage inductance (`L_leakage`) creates a voltage spike (`L_leakage * di/dt`) across the switch, adding to the voltage stress.

3. Dual Switch Forward Converter

The dual switch forward converter aims to mitigate the leakage inductance voltage kick and provide a lossless core reset.
* **Topology**: It uses two primary switches (S1, S2) and two diodes (D1, D2) in the primary side, connected across the transformer primary.
* **Operation during `DTs` (Switches ON)**:
* Both S1 and S2 are ON.
* `Vin` is applied across the primary winding.
* The secondary side operates similarly to a standard forward converter, with D1 ON and D2 OFF.
* The output inductor current (`IL`) rises.

* **Operation during `(1-D)Ts` (Switches OFF)**:
* Both S1 and S2 are OFF.
* The magnetizing current, instead of being dissipated, is returned to the input source through diodes D1 and D2. This provides a lossless core reset.
* The voltage across the primary winding becomes zero, and the flux remains constant during this period.
* The output inductor current (`IL`) falls through the secondary diode D2 (now ON).
* **Advantages**:
* Lossless core reset, improving efficiency.
* Eliminates the voltage kick on the switches caused by leakage inductance, as the energy is recycled.
* The voltage rating of the switches (S1, S2) is limited to `Vin`.
* **Disadvantages**:
* Requires two switches and two diodes in the primary side, increasing component count and complexity.
* Still suffers from limited switching frequency due to the need for core reset time.
* Core utilization remains limited to half (flux swings from 0 to `Bmax` and back to 0).
* Reverse recovery losses in the secondary diodes persist.

4. Push-Pull Converter

The push-pull converter is designed to overcome the core underutilization issue present in forward converters by driving the transformer in both directions.
* **Topology**: Features a center-tapped primary winding with two switches (S1, S2) and a center-tapped secondary winding with two diodes (D1, D2). All windings are on the same core.
* **Operation Cycle (Total `Ts`)**: The operation is typically divided into four sub-periods: `DTs/2`, `(1-D)Ts/2`, `DTs/2`, and `(1-D)Ts/2`.
* **Period 1 (`DTs/2`): S1 ON, S2 OFF**:
* `Vin` is applied across the upper half of the primary winding.
* The transformer is driven in one direction.
* On the secondary side, D1 is forward-biased, and D2 is reverse-biased.
* The voltage across the output inductor (`VL`) is `n*Vin - V0`.
* The flux in the core rises from `-Φm` towards `+Φm` with a slope of `Vin/Np`.
* The output inductor current (`IL`) rises.
* Switch S1 conducts current, while S2 is OFF.
* **Period 2 (`(1-D)Ts/2`): S1 OFF, S2 OFF**:
* No voltage is applied to the primary windings.
* The output inductor current (`IL`) distributes between D1 and D2, forward-biasing both.
* The secondary windings are effectively short-circuited, making the voltage across

the primary windings zero.
* The flux in the core remains constant at its peak value reached in Period 1.
* The voltage across the output inductor (`VL`) is `-V0`.
* `IL` falls.
* Both S1 and S2 block `Vin`.
* **Period 3 (`DTs/2`): S1 OFF, S2 ON**:
* `Vin` is applied across the lower half of the primary winding, but with reversed polarity compared to Period 1.
* The transformer is driven in the opposite direction.
* On the secondary side, D2 is forward-biased, and D1 is reverse-biased.
* The voltage across the output inductor (`VL`) is `n*Vin - V0`.
* The flux in the core falls from `+Φm` towards `-Φm` with a slope of `-Vin/Np`.
* The output inductor current (`IL`) rises.
* Switch S2 conducts current, while S1 is OFF.
* **Period 4 (`(1-D)Ts/2`): S1 OFF, S2 OFF**:
* Similar to Period 2, no voltage is applied to the primary windings.
* Both D1 and D2 are forward-biased, shorting the secondary.
* The flux in the core remains constant at its peak negative value reached in Period 3.
* The voltage across the output inductor (`VL`) is `-V0`.
* `IL` falls.
* Both S1 and S2 block `Vin`.
* **Output Voltage**: By applying volt-second balance across the output inductor over a `Ts/2` period, the output voltage is `V0 = n * D * Vin`, similar to the forward converter.
* **Advantages**:
* **Full Core Utilization**: The flux density swings from `+Bmax` to `-Bmax`, fully utilizing the magnetic core. This allows for a smaller transformer for a given power level.
* **Higher Effective Switching Frequency for Filter**: The output inductor current (`IL`) ripple frequency is twice the switching frequency (`2*f_s`). This means the output filter (L and C) can be designed for a higher frequency, leading to smaller filter components.
* **Disadvantages**:
* **High Switch Voltage Rating**: Each primary switch (S1, S2) must block `2*Vin` when OFF, requiring higher voltage-rated devices.
* **Flux Walking**: Uneven voltage drops across the switches or slight imbalances in their ON-times can lead to unequal voltage-seconds applied to the two halves of the primary winding. This causes a DC offset in the transformer flux, leading to "flux walking" where the flux progressively shifts in one direction. If uncorrected, this will eventually saturate the core, leading to catastrophic failure. Control methods (e.g.,

current sensing and duty ratio adjustment) are needed to prevent this.
* **Voltage Kick due to Leakage Inductance**: Similar to forward converters, the leakage inductance of the primary windings can cause voltage spikes on the switches during turn-off, requiring switches with even higher voltage ratings to withstand these kicks.
* **Complex Transformer**: Requires a four-winding transformer (two primary, two secondary) with a center tap, making its design and winding more complex.

Key Takeaways

- [ ] Forward converters are limited by the need for core reset time, restricting duty ratio and switching frequency.
- [ ] Basic forward converters suffer from lossy core reset and only utilize half of the transformer core's magnetic capacity.
- [ ] The demagnetizing winding in forward converters provides an active core reset but increases switch voltage stress and transformer complexity.
- [ ] Dual switch forward converters offer lossless core reset and eliminate leakage inductance voltage kicks, but still underutilize the core and require more components.
- [ ] Push-pull converters achieve full core utilization by driving the transformer in both positive and negative flux regions, allowing for smaller transformers.
- [ ] A significant disadvantage of push-pull converters is the high voltage stress (2*Vin) on the primary switches.
- [ ] Flux walking is a critical issue in push-pull converters that can lead to core saturation if not actively managed.
- [ ] Leakage inductance in isolated converters can cause damaging voltage spikes on switches during turn-off, necessitating higher voltage ratings.
- [ ] The output filter in push-pull converters can be smaller due to the inductor current ripple frequency being twice the switching frequency.

Conclusion

The exploration of isolated DC-DC converters, from the basic forward converter to its enhanced variants and the push-pull topology, highlights a continuous effort to optimize performance while managing complexity. Each converter type presents a unique set of advantages and disadvantages, primarily revolving around core utilization, switching frequency limitations, switch voltage ratings, and the impact of parasitic elements like leakage inductance. While forward converters offer simplicity, their inherent limitations in core utilization and frequency restrict their application.

Modifications like demagnetizing windings and dual-switch configurations address specific issues but introduce new trade-offs. The push-pull converter stands out for its ability to fully utilize the transformer core, leading to more compact designs, but demands careful attention to switch voltage ratings and the critical phenomenon of flux walking. The choice of converter topology is therefore a critical design decision, requiring a thorough understanding of these trade-offs to meet the specific requirements of efficiency, size, cost, and reliability in power conversion systems.